

3.3.1 Number of research papers published per teacher in the journals notified on UGC Care list during the last five years

2018

List of paper publications

S.NO	Title of the paper	Name of the author/s	Name of the journal	Page No
1	Insilico studies of compounds isolated from methanol extract of Elephantopus Scaber by GCMS against anti-arthritis protein PAD4	Antro Jennie X , Ajay Kumar M, Jenila Bejads X, Rajesh M	Journal of global trends in Pharmaceutical sciences.	7
2	A new gradient RP-HPLC method development and validation for simultaneous estimation of lamivudine, tenofovir disoproxil fumarate and Efavirenz in pharmaceutical dosage forms	KS.Sumanth , A.Srinivasarao, D.Gowrishankar	International journal of pharmaceutical, chemical and biological sciences.	8-9
3	A new stability indicating RP-HPLC method development and validation for simultaneous estimation of emtricitabine tenofovir with degradation kinetics.	KS.Sumanth , A.Srinivasarao, D.Gowrishankar	Asian Journal of Research in Chemistry	10



IN SILICO STUDIES OF COMPOUNDS ISOLATED FROM METHANOL EXTRACT
OF ELEPHANTOPUS SCABER BY GCMS AGAINST ANTI-ARTHRITIC PROTEIN
PAD4

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ARTICLE INFO

ABSTRACT

Key Words

Elephantopus scaber
PADS
In Silico Studies
Anti-arthritis protein
PAD4



Protein Arginine Deiminase4 is an enzyme responsible for the citrullination of antigens in rheumatoid arthritis. PADS are calcium dependent enzymes because PADS rely on increased concentration of calcium to citrullinate peptidyl arginine. PADS are not normally active until stimulated with calcium. C1 amidase, F4 amidase are the most potent PAD inhibitor. The present study, evaluated the binding of certain GCMS derived compounds of the methanol extract of "*Elephantopus scaber*" against the Antiarthritis protein PAD-4. Docking scores were compared with their binding energy as well as affinity using the Schrodinger, 2015.1. The docking or Glide score of the above six compounds are -7.9, -5.2, -4.87, -3.77, -3.63, -3.33 kcal/mol respectively and the glide energy were -33.83, 24.80, -30.06, -34.67, -30.38 and -16.29 kcal/mol. The other properties such as Lipinski rule, absorption, distribution, metabolism, excretion (ADME), toxicity predictions, logP values and % of human oral absorption were analyzed using high-throughput screening. The results of our study showed that out of 9 compounds, six compounds of *E. scaber* may act as good inhibitors for arthritis and the compounds can be re-designed for better antiarthritis activity for synthesis.

INTRODUCTION:

Elephantopus scaber Linn. Commonly known as Elephant's foot, (Asteraceae) is a scabrescent aromatic herb distributed in the moist deciduous forests of central Western Ghats. According to the traditional claims, the roots were used as an antipyretic, cardiotonic, dysuria, diarrhea, dysentery, stomach pain, and diuretic. The

leaf extract had already been evaluated for diuretic, anti-inflammatory, hepatoprotective effect, and also treatment for eczema and ulcers. *E. scaber* is used in Chinese medicine for the treatment of some types of cancer. Bioactivity screening of the extracts and their compounds had shown that *E. scaber* possess wound healing, anti-venom,

A NEW GRADIENT RP-HPLC METHOD DEVELOPMENT AND VALIDATION FOR SIMULTANEOUS ESTIMATION OF LAMIVUDINE, TENOFOVIR DISOPROXIL FUMARATE AND EFFAVIRENZ IN PHARMACEUTICAL DOSAGE FORMS

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ABSTRACT

A new, simple, rapid and accurate gradient RP-HPLC method was developed for the estimation of Lamivudine (LAM), Tenofovir disoproxil fumarate (TDF) and Effavirenz (EFA) in pharmaceutical dosage forms and validated. The HPLC method was developed on SHISEIDO C18 column (250 x 4.6 mm i.d, 5 μ) using Acetonitrile: 50mM phosphate buffer (pH 5.0) at 256 nm. The retention times for Lamivudine, Tenofovir disoproxil fumarate and Effavirenz were found to be 2.4, 4.1 and 12.2 min respectively. Linearity was established in the range of 5-25 g/ml, 5-25 g/ml and 10-50 g/ml for LAM, TDF and EFA respectively. The coefficients of regression (R²) values were found to be 0.997, 0.995 and 0.997 for LAM, TDF and EFA respectively. The method was precise with %RSD

< 2 for both intraday and interday precision. The accuracy of the method was performed over three levels of concentration and the recovery was in the range of 98-102%. The method was successfully applied for quantifying these drugs in marketed dosage forms.

INTRODUCTION

Lamivudine (LAM) is a nucleoside reverse transcriptase inhibitor and cytidine analog. It is used to treat human immune deficiency virus type 1 (HIV-1) and Hepatitis B (HBV). Chemically it is 4 - amino-1- [(2R, 5S) 2 - (hydroxymethyl) - 1, 3 oxathiolan 5 yl] - 1, 2 dihydropyrimidin 2 one (Fig. 1). Tenofovir disoproxil fumarate (TDF) is a fumaric acid salt of the bis- isopropoxy carbonyl oxymethyl ester derivative of tenofovir. Chemically it is 9-[(R)-2[[bis[[(isopropoxycarbonyl) oxy] methoxy] phosphinyl] methoxy] propyl] adenine fumarate (Fig. 2). Effavirenz (EFA) is a non-nucleotide reverse transcriptase inhibitor (NNRT) and is used as a part of highly active anti-retroviral therapy (HAART) for the treatment of the Human Immuno Deficiency virus (HIV Type-1).

Chemically it is (4*S*)-6-chloro-4-(2- cyclopropylethynyl)-4-(trifluoromethyl)-2,4- dihydro-1*H*-3,1-benzoxazin-2-one¹⁻³ (Fig. 3).

The present study is to develop a gradient RP-HPLC method for LAM, TDF and EFA. The Literature survey reports different analytical methods for LAM, TDF and EFA based on UV⁴⁻⁵, HPLC⁶⁻¹⁴, stability indicating HPLC¹⁵⁻¹⁹, Fluorimetry²⁰, Chiral chromatography²¹, LC-MS^{22,23} were reported. However there were few methods reported for the simultaneous estimation of LAM, TDF and EFA by gradient chromatography, the present aim is to develop a more precise, accurate, simple and gradient RP-HPLC method for the estimation of LAM, TDF and EFA. The molar absorptivity of LAM, TDF and EFA were found maximum at



RESEARCH ARTICLE

A New Stability Indicating RP-HPLC Method Development and Validation for Simultaneous Estimation of Emtricitabine and Tenofovir with Degradation Kinetics.

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ABSTRACT:

A new, simple, rapid and stability-indicating RP-HPLC method was developed for the estimation of Emtricitabine (EMT) and Tenofovir disoproxil fumarate (TDF) in pharmaceutical dosage forms and validated. The HPLC method was developed on SHISEIDO C18 column (250 x 4.6 mm i.d, 5 μ) using methanol: 20mM phosphate buffer (pH 5.0) in the ratio of 35:65 v/v at 271 nm. The retention times for EMT and TDF was found to be 2.79 and 4.91 min respectively. Linearity was established in the range of 0.5-1.5 g/ml and 0.75-2.0 g/ml for EMT and TDF respectively. The method was precise with %RSD < 2 for both intraday and interday precision. The accuracy of the method was performed over three levels of concentration and the recovery was in the range of 98-102%. The drugs were individually subjected to forced degradation (thermal, photolytic, hydrolytic, and oxidative stress conditions) studies for seven days. EMT showed maximum degradation in acidic medium and TDF showed maximum degradation in basic medium. The method was successfully applied for quantifying the drugs in marketed dosage forms.

KEYWORDS: RP-HPLC, Tenofovir disoproxil fumarate, Emtricitabine, Phosphate buffer, Forced degradation.

1. INTRODUCTION:

Emtricitabine (EMT) is a nucleoside reverse transcriptase inhibitor (NRTI). Chemically it is 5-fluoro- 1-(2R, 5S)-[2-(hydroxymethyl)-1, 3-oxathiolan-5-yl] cytosine (Fig. 1). EMT is the (-) enantiomer of thio analog of cytidine which differs from other cytidine analogs, in that it has fluorine in the 5th position. EMT is an antiviral agent used for the prevention of perinatal HIV-1 reverse transcriptase^[1].

It is also active against Hepatitis B virus^[2]. Tenofovir disoproxil fumarate (TDF) is a fumaric acid salt of the bis-isopropoxy carbonyl oxymethyl ester derivative of tenofovir. TDF is a nucleotide reverse transcriptase inhibitor. TDF is converted to Tenofovir, an acyclic nucleoside Phosphonate (Nucleotide) analog of adenosine 5'-Monophosphate. Chemically it is 9-[(R)-2-[bis[(isopropoxycarbonyl) oxy] methoxy] phosphinyl] methoxy] propyl] adenine fumarate^[3] (Fig. 2).

The present study is to develop a stability indicating RP-HPLC method for EMT and TDF. The objective of the study is to subject the drugs for acid, base, peroxide

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